

FANUC Robot **series**

R-J3/R-J3*i*B/R-30*i*A/R-30*i*B Plus/R-30*i*B Mate Plus CONTROLLER 5.20 and later

High Speed Digital Input Module (HSI) OPTIONS MANUAL

MARO3HSIP12901E REV F

ORIGINAL INSTRUCTIONS

Thank you very much for purchasing a FANUC robot.

Before using the robot, be sure to read the, *FANUC Robot series SAFETY HANDBOOK (B-80687EN)* and understand its contents.

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In this manual, we endeavor to include all pertinent matters. There are, however, a very large number of operations that must not or cannot be performed. Please assume that any operations that are not explicitly described as being possible are not possible.

SAFETY PRECAUTIONS

This chapter describes the precautions which must be followed to enable the safe use of the robot. Before using the robot, be sure to read this chapter thoroughly.

For detailed functions of the robot operation, read the relevant operator's manual to understand fully its specification.

For the safety of the operator and the system, follow all safety precautions when operating a robot and its peripheral equipment installed in a work cell. For safe use of FANUC robots, you must read and follow the instructions in the *FANUC Robot series SAFETY HANDBOOK (B-80687EN)*.

PERSONNEL

Personnel can be classified as follows:

Operator:

- Turns the robot controller power on/off
- Starts the robot program from the operator panel

Programmer or Teaching operator:

- Operates the robot
- Teaches the robot inside the safeguarded space

Maintenance technician:

- Operates the robot
- Teaches the robot inside the safeguarded space
- Performs maintenance (repair, adjustment, replacement)

The operator is not allowed to work in the safeguarded space.

The programmer or teaching operator and maintenance technician are allowed to work in the safeguarded space. Work carried out in the safeguarded space include transportation, installation, teaching, adjustment, and maintenance.

To work inside the safeguarded space, the person must be trained on proper robot operation.

Table s-1 lists the work outside the safeguarded space. In this table, the symbol “○” means the work is allowed to be carried out by the specified personnel.

Table s-1 Work Performed Outside the Safeguarded Space

	Operator	Programmer or Teaching Operator	Maintenance Technician
Turn power ON/OFF to Robot controller	○	○	○
Select operating mode (AUTO, T1, T2)		○	○
Select remote/local mode		○	○
Select robot program with teach pendant		○	○
Select robot program with external device		○	○

Start robot program with operator's panel	○	○	○
Start robot program with teach pendant		○	○
Reset alarm with operator's panel		○	○
Reset alarm with teach pendant		○	○
Set data on teach pendant		○	○
Teaching with teach pendant		○	○
Emergency stop with operator's panel	○	○	○
Emergency stop with teach pendant	○	○	○
Operator Panel maintenance			○
Teach Pendant maintenance			○

During robot operation, programming and maintenance, the operator, programmer, teaching operator and maintenance engineer take care of their own safety using at least the following safety protectors:

- Use clothes, uniform, overall adequate for the work
- Safety shoes
- Helmet

DEFINITION OF SAFETY NOTATIONS

To ensure the safety of workers and prevent damage to the machine, this manual indicates each precaution on safety according to its severity. Read the contents of each **Warning** and **Caution** before attempting to use the robot.

WARNING

Indicates a hazard resulting in the death or serious injury of the user could occur if he or she fails to follow the approved procedure.

CAUTION

Indicates a hazard resulting in the injury of the user or damage to equipment could occur if the user fails to follow the approved procedure.

Note

Indicates a supplementary explanation not related to any WARNING or CAUTION.

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1 OVERVIEW

The High Speed Digital Input Module (HSI) option consists of an electronic module and software interface. The software interface captures and displays the pulse timing, count, and frequency of the two available input channels. A typical application for this function is fluid flow measurement where the pulse rate is proportional to fluid flow rate. See [Figure 1 \(b\)](#).

Note

The new HSI module (EE-8836-703-000) replaces the now obsolete HSI modules (EE-1063-300 and EE-1063-400). The differences between the EE-8836-703-000 and the obsolete modules are explained in this manual. See [Table 2.2](#) for a comparison between the old and the new.

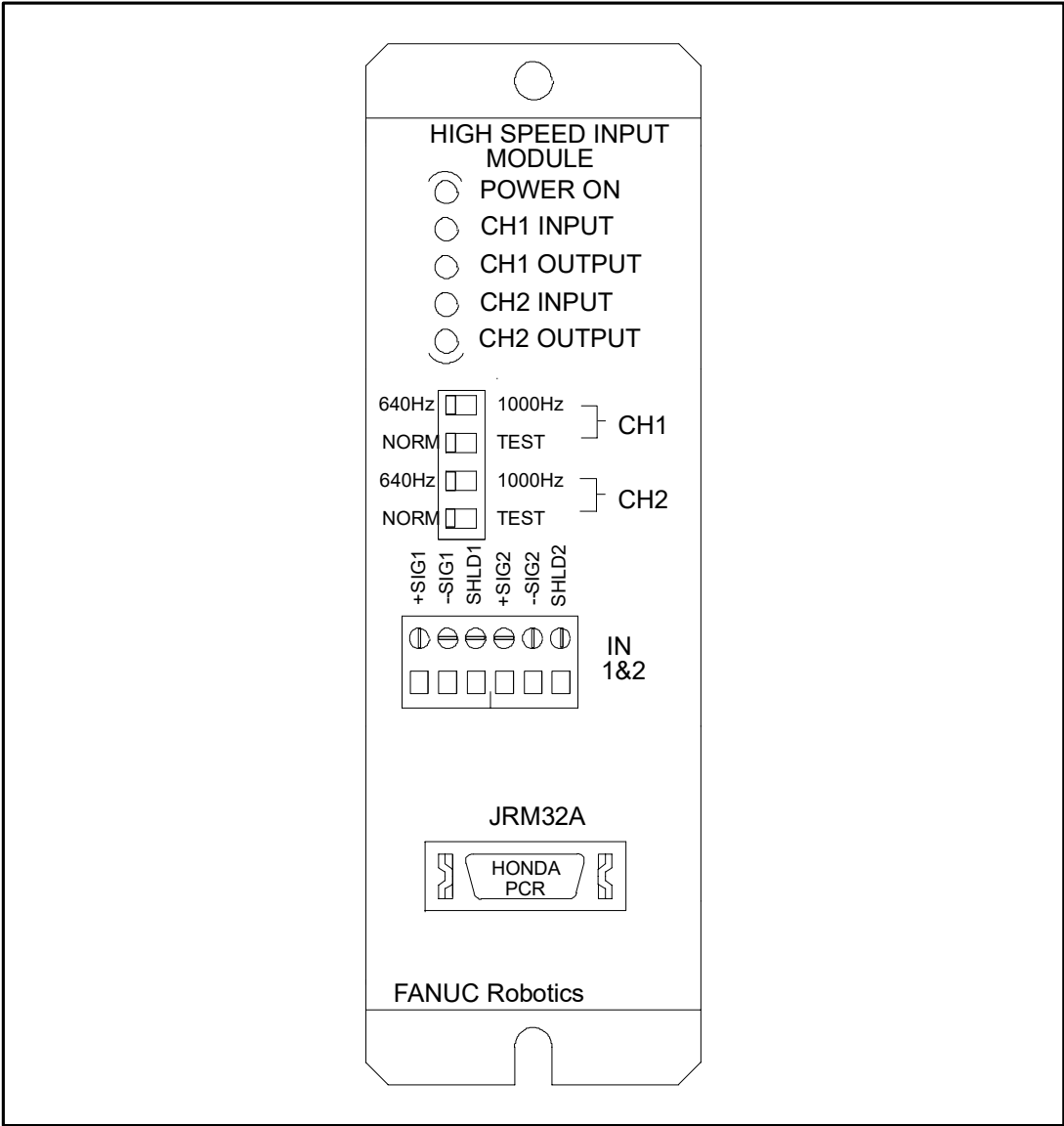


Figure 1 (a) High Speed Digital Input Module (HSI) - EE-1063-300, EE-1063-400

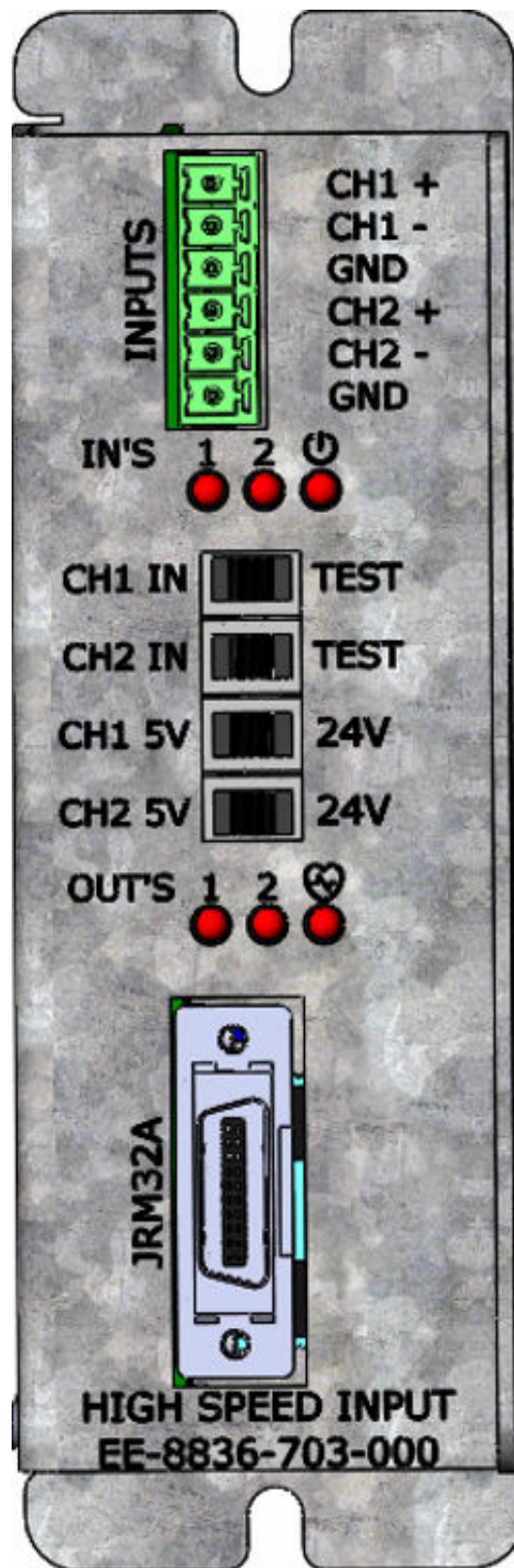


Figure 1 (b) High Speed Digital Input Module (HSI) - EE-8836-703-000

This interface:

- Provides two high speed inputs
- Includes optical signal isolation and maximum frequency filter circuits for noise rejection.
- Includes an interface to the High Speed Digital Inputs (HSI) for the R-30iA, R-30iB/R-30iB Mate, R-30iB Plus/R-30iB Mate Plus controller.

- Measures relative frequency based on pulse timing per input channel.
- Measures total pulse count per input channel.
- Is compatible with any application tool for the robot controller.
- Does not require Model A or Model B I/O components.
- Directly connects to the robot controller mainboard.

The module was designed to function with the following device types:

- Positive Displacement (PD) Flow Meters
- Coriolis Flow Meters
- Mass Flow Meters
- Proximity Sensors
- Fiber Optic/Photoelectric Sensors

2 SPECIFICATIONS

2.1 GENERAL SPECIFICATIONS

The HSI Module is used to capture high speed control signals from industrial sensors and devices. This component operates within the 5-24VDC signal range and converts to a TTL logic level (5V). [Table 2.1](#) shows the general specifications of the HSI module. For more details, refer to [Table 2.2](#).

Table 2.1 Controller Limited Specifications

Parameter	Value
Minimum Frequency	1Hz
Resolution	1mHz
Accuracy	0.5% (1–400Hz) 0.9% (400 - 1000Hz)
Operating temperature range	0 - 55°C
Response time of hardware	—0.15uS
External Power Required	None

2.2 MODULE SPECIFICATIONS

Two input channel are available. Refer to [Figure 2.2 \(a\)](#) and [Figure 2.2 \(b\)](#) for the equivalent circuit details. The input signals are filtered to mitigate electrical interference and to prevent system overload associated with high frequency inputs. The filters are composed in a monostable multivibrator circuit that inherently limits the number of transitions observed in the software. In this manner, false signals are rejected. The rising edge of the optoisolator triggers a signal sent to the HSI input(s). The monostable extends the rising edge pulse by time TL.

Refer to [Table 2.2](#) and for detailed information.

Table 2.2 HSI Module Specifications

Parameter	EE-8836-703-000	EE-1063-300 (Obsolete)	EE-1063-400 (Obsolete)
Mounting Slot	5mm	5mm	5mm
Mounting Width	130mm	130mm	130mm
Width	45mm	45mm	45mm
Height	140mm	140mm	140mm
Depth	80mm	58mm + Cable	58mm + Cable
TL {F1}	1.02ms	1.563ms	1.563ms
TL {F2}		1.02ms	0.833ms
Period {T} Tolerance	11%	12%	12%
Resistance Input {5V}	560Ω	-	470Ω

Parameter	EE-8836-703-000	EE-1063-300 (Obsolete)	EE-1063-400 (Obsolete)
Resistance Input {24V}	2260Ω	1800Ω	-
On MIN Input {5V}	4.1V +	-	3.54V
On MIN Input {24V}	5.6V +	8.1V	-
VIN MAX Input {5V}	14.8V *	-	5.5V
VIN MAX Input {24V}	26.7V *	24.5V	
OFF, Current MAX	0.3mA	0.3mA	0.3mA
OFF, Voltage MAX	2.62V	2V	1.75V
Response Time (1mA)	20uSec	-	-
Response Time (4mA)	3uSec	-	-
Test Frequency	21-27Hz	17-23Hz	17-23Hz
Isolation Voltage	2500V	2500V	2500V
MAX Frequency = 1/TL	872Hz-1190Hz	-	-
# Maximum frequency is determined by RC tolerance. Frequencies > 1/TL will hold the output LOW, and will read as 0 Hz. * Maximums are based of power rating of resistors. + "ON" Minimums are based of 1mA of current through diodes. "OFF" Minimums are based off 0.3mA. Input conditions close to operations points could lead to varying results			

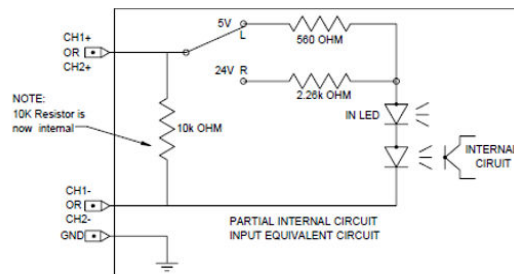


Figure 2.2 (a) Equivalent Input Circuit - EE-8836-703-000

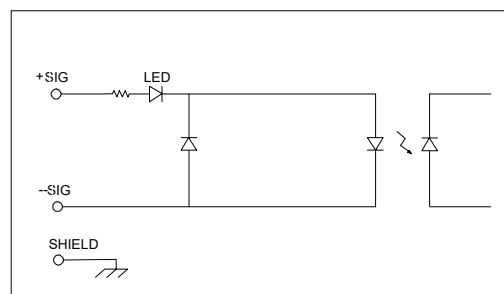


Figure 2.2 (b) Equivalent Input Circuit - EE-1063-300, EE-1063-400 (Obsolete)

3 FREQUENCY INTERFACE

3.1 SOFTWARE

The HSI Interface Option is included with R548: Frequency Interface controller software. [Table 3.1](#) shows the respective system variables available for both High Frequency input channels.

Table 3.1 Information on Channel System Variables

System Variables for Reading Pulse Count and Frequency		
Channel 1	\$fqint_res[1].\$total_pulse	Number of pulses received. Can be reset to 0
	\$fqint_res[1].\$freq_read	Frequency reading. (1000 = 1.00 Hz)
Channel 2	\$fqint_res[2].\$total_pulse	Number of pulses received. Can be reset to 0.
	\$fqint_res[2].\$freq_read	Frequency reading. (1000 = 1.00 Hz)

3.1.1 POSITION SNAP AND MAP

3.1.1.1 OVERVIEW

The Position Snap and Map feature, available in controller software version 6.40 and later, records the current Robot Position in encoder counts when a single High Speed Digital Input (H.D.I.) is turned on. In a system with a robot-mounted vision system camera or a measurement sensor, a signal is provided to both the camera and the robot controller. This signal causes the robot to snap its current position at the same instant that the image is snapped. Later, additional software acquires and relates the robot positional data with the image taken - thus a map of the robot's position. This functions with the robot moving at reasonable speeds, thus increasing cycle time and accuracy of the system.

Refer to [Section 3.1.1.5, EXAMPLE PROGRAM](#) for an example KAREL program.

The functionality provides the following:

1. When the input transitions from false to true then record the current real time encoder values. Currently this input must be one of the High Speed Digital Inputs.
2. The recorded position has the following characteristics:
 - A single position will be stored into a system variable. This is the absolute encoder pulse count reading of the current robot position, up to six axes. The axes angles are also provided. This is the same as the current joint position.
 - The response from the transition of the HDI input to the reading of the position is at a maximum of 1 millisecond, and averages .5 milliseconds.
 - The position can optionally include the Extended Axes that are a part of Group No. 1.
 - Only a single Motion Group, Group No. 1 is supported.
 - The current value of \$FAST_CLOCK (4 ms Ticks) is included in the system variable.
 - Any axis that is not used by the robot is set to a value of 0.

Note

Position Snap and Map is not compatible with the High Speed Line Tracking feature because both features use HDI number 1.

3.1.1.2 INSTALLATION

To use this feature you must install the Position Snap & Map (WMAP) software option, and install the HDI pulse cable. Refer to the Software Installation Manual for information on installing software options. The hardware module and cable are described in the other chapters of this manual. To install this feature there is a software option to be loaded and there is a cable or hardware option needed. The software option is called "Position Snap & Map (WMAP)." The software option can be installed before or after the hardware installation. The software option should be installed according to the instructions in the Software Installation Manual. The hardware to be installed is the High Speed Input module that is documented in [Chapter 4, MODULE INSTALLATION](#)

Note

If the robot is a six axes robot and HDI Number 1 will be used then there should not need to be any changes made to the system variable that configures this option. Otherwise, review and modify the elements of the "\$WMCFG" system variable. NOTE: After each power up the KAREL or teach pendant program that reads the position data from the system variables must set \$record to TRUE.

3.1.1.3 SYSTEM VARIABLES

The system variables used in Position Snap and Map are:

- \$WMCFG:
 - \$WMCFG.\$ENABLE
 - \$WMCFG.\$IN_IO_TYP
 - \$WMCFG.\$RECORD
 - \$WMCFG.\$IN_IO_NO
 - \$WMCFG.\$NUM_AXES
 - \$WMCFG.\$DEBUG
- \$WMDATA:
 - \$WMDATA.\$MACHINE_PLS
 - \$WMDATA.\$LAST_F_CLOCK
 - \$WMDATA.\$REC_POINT

Note

Refer to the *Software Reference Manual* for more information on each system variable.

3.1.1.4 ERROR MESSAGES

Three errors can occur during power up initialization with this function:

- WMAP-001 STOP WM error allocating data
- WMAP-002 STOP HSDI system call error
- WMAP-002 STOP Invalid \$wmcfg setting

Note

Refer to the *Error Code Manual* for cause and remedy information on each error message.

3.1.1.5 EXAMPLE PROGRAM

The following is an example program for acquiring the position data from the *Position Snap* option. This program can be run as is to confirm the installation of the hardware and software. The key and most common component is the condition handler with the clause:

```
"WHEN ( $wmdata.$lst_f_clock <> prv_f_clock ) DO"
```

Translation of this program will require a copy of the *Environment* file for the *Position Snap* option. This file is named *wmdef.ev*.

```
file for the "Position Snap" option.
This file is named wmdef.ev.
-- Program: wmapexa - Position Snap & Map example program
--
-- Description: This program provides an example for
-- reading the Position Snap data system variables
--
-- Language: KAREL
--
-- Source File: wmapexa.kl 28-JUN-2004
--
-----
--
PROGRAM wmapexa
-----
--
%COMMENT='WMAP Example'
%STACKSIZE = 400
%NOLOCKGROUP
%NOPAUSE = ERROR + COMMAND + TPENABLE
%NOABORT = ERROR + COMMAND
%NOBUSYLAMP
%ENVIRONMENT sysdef -- system variables
%ENVIRONMENT wmdef
CONST
ch_test1 = 1 -- Condition handler for tests
ch_test2 = 2
max_q = 100 -- Number of samples that can be saved
TYPE
result_str from wmapexa = structure
axis_ang : ARRAY[9] OF REAL
lst_f_clock : integer
spare : integer
endstructure
VAR
cont_test : BOOLEAN
q : INTEGER
q_cnt : INTEGER
prv_f_clock : INTEGER
map_pos : XYZWPR
temp_jpos : JOINTPOS
status : INTEGER
```

```

test_count IN CMOS : INTEGER
mpos_results : ARRAY[MAX_Q] of result_str
null_result : result_str
-----
--
ROUTINE new_result
BEGIN
-- Save the results or do other operations on the new data
mpos_results[q].axis_ang = $wmdata.$rec_point
mpos_results[q].lst_f_clock = prv_f_clock
q=q+1
IF (q > max_q) THEN
q=1
ENDIF
ENABLE CONDITION[ch_test1]
END new_result
BEGIN
q =1
q_cnt = 0
status = 0
prv_f_clock = 0
cont_test = true
if UNINIT(test_count) then
test_count = 30
endif
WRITE (CR)
WRITE (CR)
WRITE (' Wmapexa Start', CR)
DELAY(1000) --
null_result.lst_f_clock = 0
null_result.spare = 0
map_pos = $MOR_GRP[1].$NILPOS
q=1
-- Intial. of the whole axes angle array
while (q <= 9) do
null_result.axis_ang[q] = 0
q = q+1
endwhile
q=1
-- Complete the intial. of the whole results array
while (q <= MAX_Q) do
-- Copy whole structure in one statement:
mpos_results[q] = null_result
q = q+1
endwhile
prv_f_clock = $wmdata.$lst_f_clock
q=1
-- Condition handler to gather the data
CONDITION[ch_test1]:
WHEN ( $wmdata.$lst_f_clock <> prv_f_clock ) DO
prv_f_clock = $wmdata.$lst_f_clock
new_result
-- In the routine:
-- ENABLE CONDITION[ch_test1]
ENDCONDITION
q=1
ENABLE CONDITION[ch_test1]
delay (500)
if (q = 1) then
WRITE ('Trigger the HDI input', CR)
endif

```

```

WHILE (cont_test) AND (q <= 1) DO
  DELAY(1000)
ENDWHILE
if (q > 1) then
  WRITE ('First HDI - OK', CR)
endif
WHILE (cont_test) AND (q < test_count) DO
  DELAY(100)
ENDWHILE
DISABLE CONDITION[ch_test1]
-- If there is an axis angle, convert to XYZWPR Position type
if (q > 1) then
  -- convert an axis angle, convert to joint Position type
  cnv_rel_jpos(mpos_results[1].axis_ang, temp_jpos, status)
  -- convert joint pos. to XYZWPR Position type
  map_pos = temp_jpos
endif
WRITE ('Program wmapexa completed', CR)
END wmapexa

```

3.2 FREQUENCY READING

The Frequency Reading is supplied as an integer in units of “x1000 Hz”. For example a value of 12.670 indicates a frequency of 12670 Hz. The maximum frequency is limited by the electronic interface and the accuracy. The maximum frequency is 1000 Hertz. This provides 0.1% resolution at 1 Hertz.

Changes to the Total Pulses Count value will indicate that there is a new Frequency Reading available.

Optionally, this frequency reading is smoothed or filtered using a sampling algorithm as follows:

Average Rate New = ([{Sample Amount--1} * Average Rate Old] + Instantaneous Rate New)

Note

For more information on the system variables for the Frequency Interface, including "Sample Amount", refer to the *Controller Software Reference Manual*, Version 6.20 or later.

This filtering is enabled and adjusted by setting the “Sample Amount” system variable.

The Frequency Reading can be accessed at the following system variables: **\$fqint_res[1].\$freq_read** and **\$fqint_res[2].\$freq_read**

3.3 TOTAL PULSE COUNT

The Total Pulse Count value is the running sum of incoming pulse signals. It is updated every time a new rising edge is detected. The count is updated normally until the register overflows, at which time the count is wrapped to zero and continues.

A reset of the total pulses is accomplished in the software by setting the Total Pulses value to 0. This value can also be set to a non-zero value if needed.

The range of the Total Pulse Count value is limited to 4 bytes or a maximum value of 2147483647.

Pulse counts are lost if the maximum frequency is exceeded. An alarm is available to detect this issue but must be configured and enabled by the user. Contact FANUC America Corporation for additional information.

The Total Pulse Count can be accessed at the following system variables: \$fqint_res[1].\$total_pulse and \$fqint_res[2].\$total_pulse

4 MODULE INSTALLATION

4.1 MODULE INSTALLATION

The following section provides information on the installation and setup of the High Speed Digital Input Module. See [Figure 4.1 \(a\)](#), [Figure 4.1 \(b\)](#), and [Figure 4.1 \(c\)](#) for typical installations. A representative layout is provided to show Intrinsically Safe Barriers for connected devices operating in hazardous location.

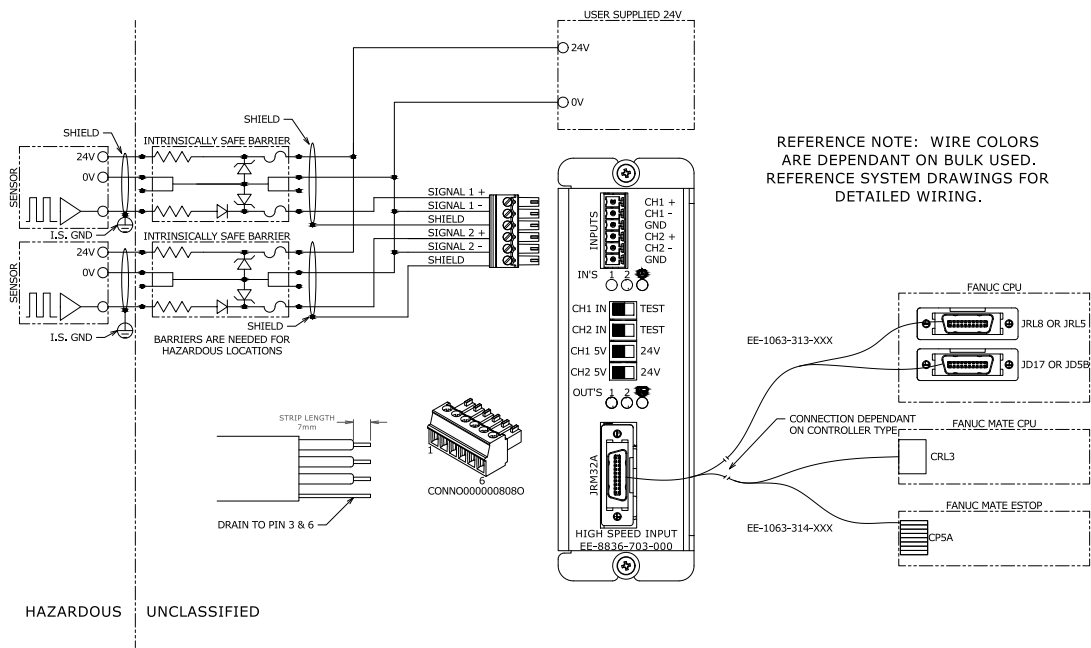


Figure 4.1 (a) Installation Block Diagram - EE-8836-703-000

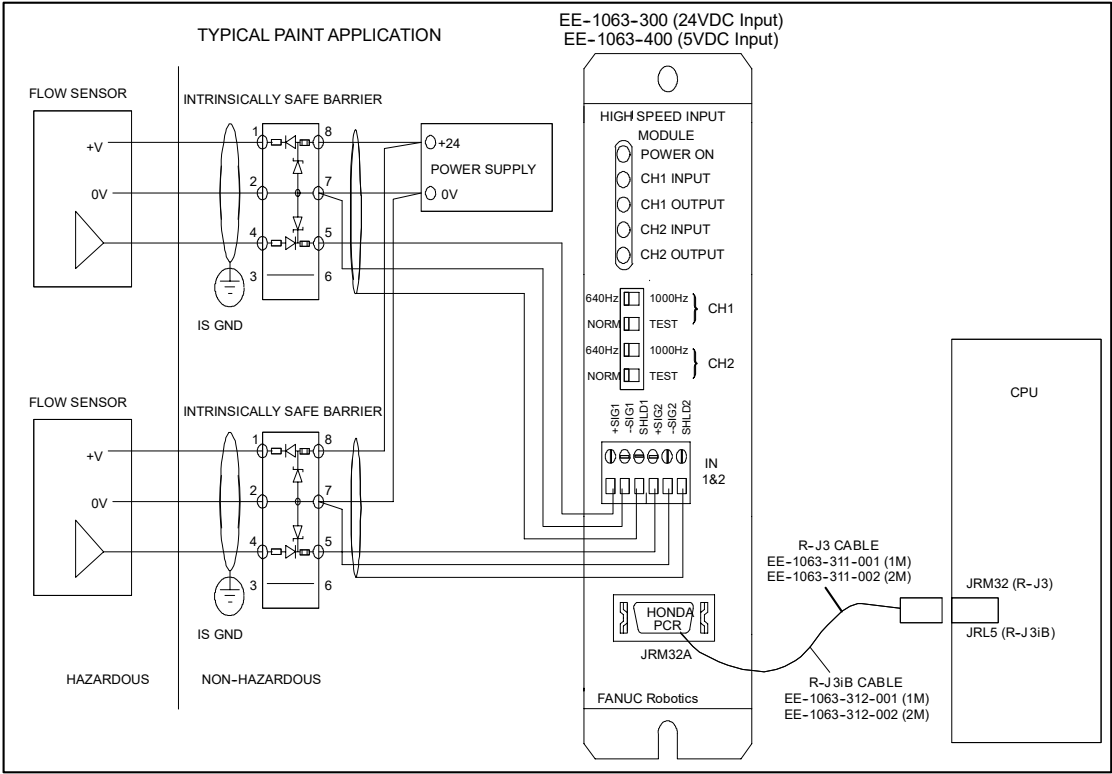


Figure 4.1 (b) Installation Block Diagram - EE-1063-300, EE-1063-400 (Obsolete)

TYPICAL HOOKUP DIAGRAMS

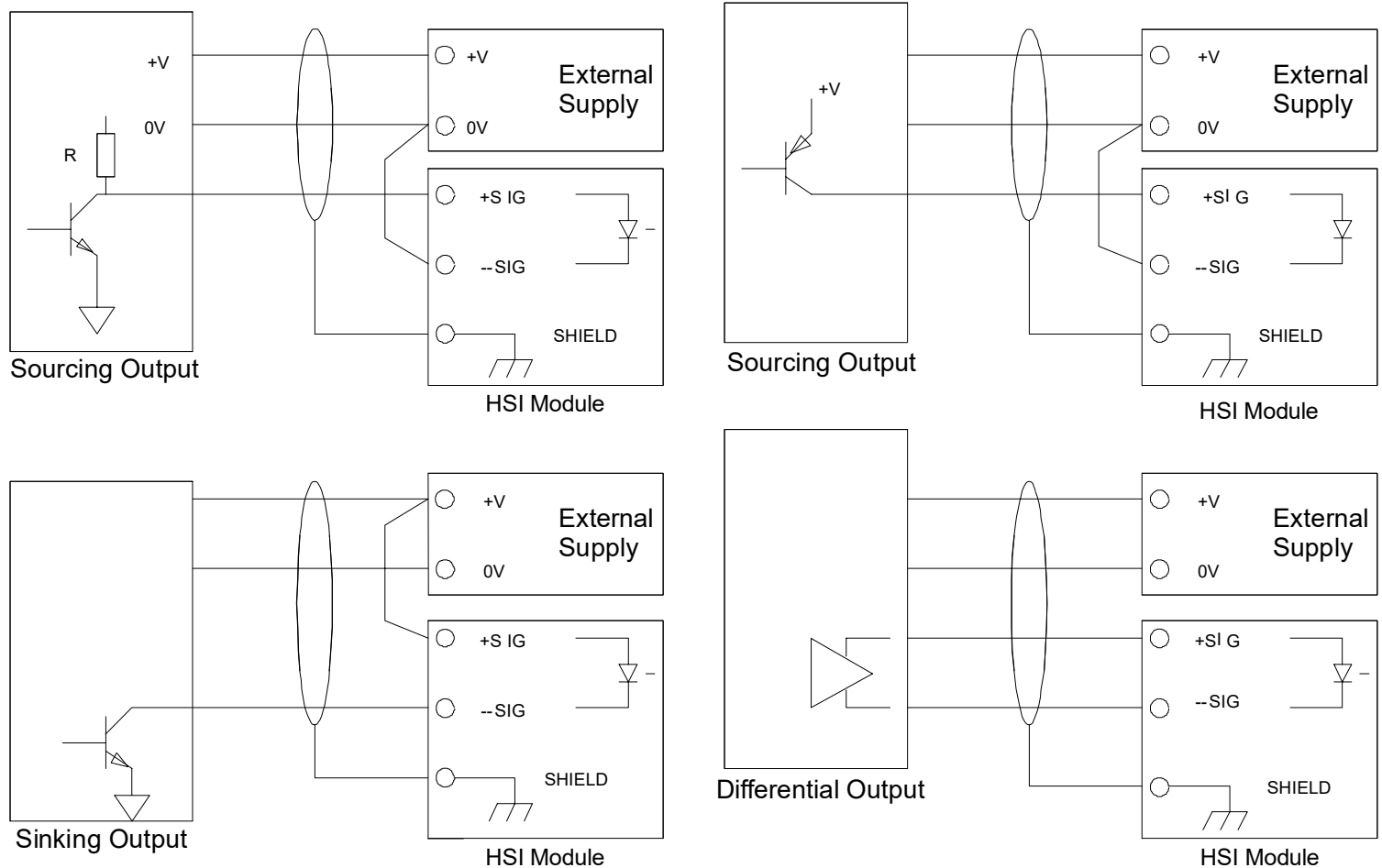


Figure 4.1 (c) Typical Connection Block Diagram

4.1.1 HSI MODULE INSTALLATION AND SETUP

Procedure

1. Mount the HSI module. The footprint of the module which describes the mounting provision is shown in [Figure 4.1.1](#). The unit requires two M4 screws and washers located on the top and bottom of the chassis.

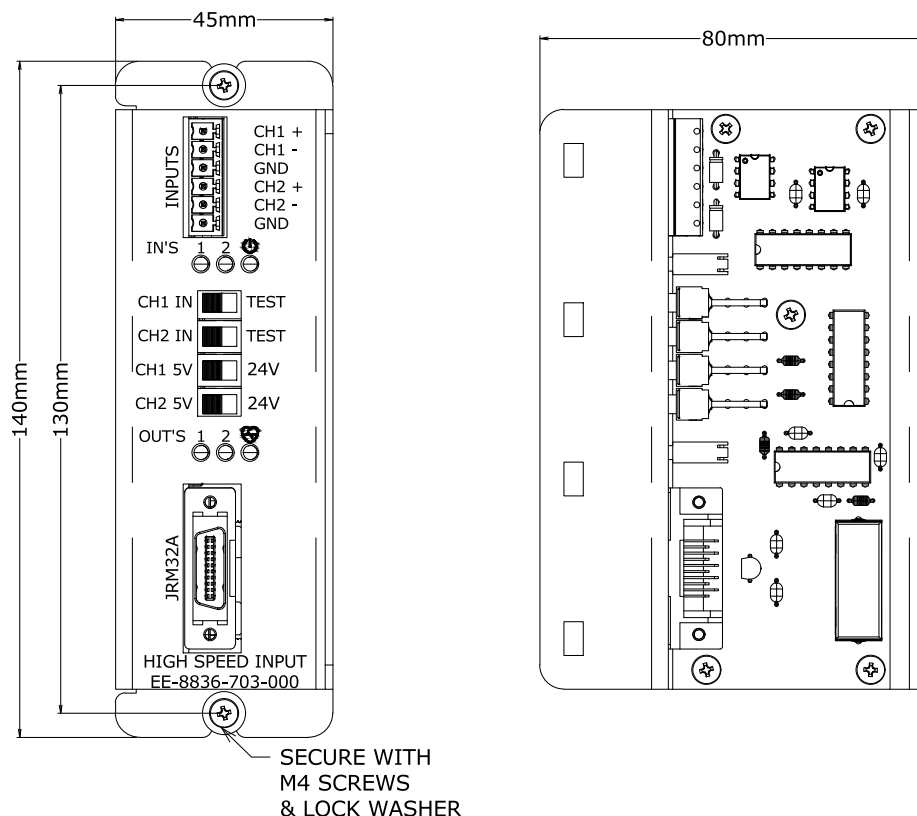


Figure 4.1.1 HSI Module Mounting Dimensions

2. Connect the sensor. Inputs from the sensors go to a Phoenix terminal block and an external power source. Connect the CHx+, CHx-, and cable shield to the terminals.
3. Connect the HSI to the CPU. The connector used is a PCR (mini Honda) connector. See [Figure 4.1 \(a\)](#) and [Figure 4.1 \(b\)](#) respectively.
 - For R-J3iB, R-30iA, R-30iB and R-30iB Mate: Connect EE-1063-312-001/002 to the JRL5 port on the Main CPU board.
 - For R-30iB Plus: Connect EE-1063-313-XXX to the JRL8 & (JD17 or JD5B) port on the Main CPU board.
 - For R-30iB Mate Plus: Connect EE-1063-314-XXX to CRL3 port on the Main CPU board, and CP5A on the E-Stop board.
4. Set the switches. Set the Channel 1 and Channel 2 switches. Select 5V or 24V based on the application and the type of sensor. Refer to [Section 5.2, SWITCH DESCRIPTIONS](#). Set the IN/TEST switch to IN.
5. If the Frequency Interface option is being used, then on the teach pendant set the Channel 1 system variable to “\$FQINT_SETUP[1].\$FREQ_ENB = TRUE”. For Channel 2, set the system variable to “\$FQINT_SETUP[2].\$FREQ_ENB = TRUE”.

Note

Power Cycle is needed after enable or software will not function.

6. Turn the controller power off and then ON.

4.2 PART NUMBERS

This section provides part numbers for the High Speed Digital Input Module (HSI).

Table 4.2 Part Numbers for the HSI Module

Part Number	Description
EE-8836-703-000	HSI Module (24V or 5V)
EE-1063-300	HSI Module (24V Input)
EE-1063-400	HSI Module (5V Input)
TERM1850709 (Obsolete)	Input Signal Connector (Phoenix)
CONNO000000808O	Input Signal Connector (Phoenix)
EE-1063-312-001,-002	R-J3iB and R-30iA Cable
EE-1063-313-XXX	R30iB Plus Cable
EE-1063-314-XXX	R30iB Mate Plus Cable

5 OPERATION

The following section describes HSI Module operation. It also provides details the functions of the switches and LEDs positioned on the face of the HSI Module.

5.1 HSI MODULE OPERATION

Procedure

1. Turn ON the controller. The HSI Module boot with the controller. Check that the *POWER ON* LED is illuminated and the *HEARTBEAT* LED is flashing. See [Section 5.3, LED DESCRIPTIONS AND TROUBLESHOOTING INFORMATION](#) for more details.
2. Test the module's functionality. Front Panel Dip Switches can be used to test connections to the CPU without the need of an external sensor or device. See [Section 5.2, SWITCH DESCRIPTIONS](#) for more details. When TEST mode is selected, the respective *OUT'S* LED will light if the hardware is working correctly. The brightness will be a fraction of the heartbeat LED. The *IN'S* LED will not illuminate unless an actual input is being received. Using the teach pendant, check the system variables shown in the Output section to see if the controller software is correctly receiving the test input. When operation is confirmed, set the DIP switches to IN mode.

Note

If the switch is maintained in the test mode, the HSI will not provide readings of the frequency input.

5.2 SWITCH DESCRIPTIONS

See switch details for the HSI Modules below:

Table 5.2 (a) Modular HSI Switches - EE-8836-703-000

Switch Settings	Description
CHx IN / TEST	IN: Set in this position for normal operation.
	TEST: Set when checking for proper hardware operation. OUT'S x LED will illuminate if the hardware is working correctly (The IN'S LED will not illuminate unless an actual input is being received). Check System Variables detailed in Section 3.1, SOFTWARE to verify operation of the HSI module.
	Note If the <i>Threshold</i> system variable is more than 17hz then the frequency readings will be zero. The frequency reading is typically 24Hz ±2.
CHx 5V / 24V	24V: For use with 24V input devices.
	5V: For use with 5V input devices. See Table 2.2 for more details.

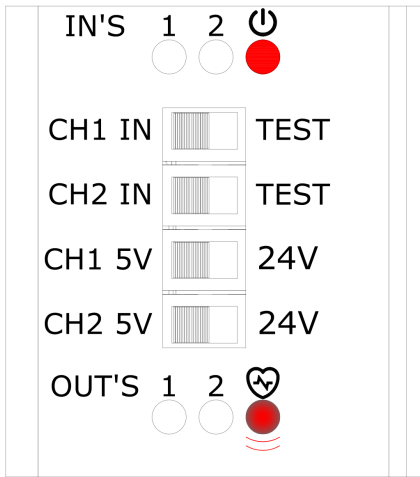
Table 5.2 (b) Modular HSI Switches - EE-1063-300, EE-1063-400 (Obsolete)

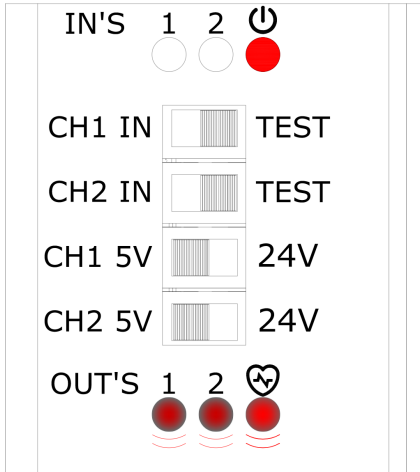
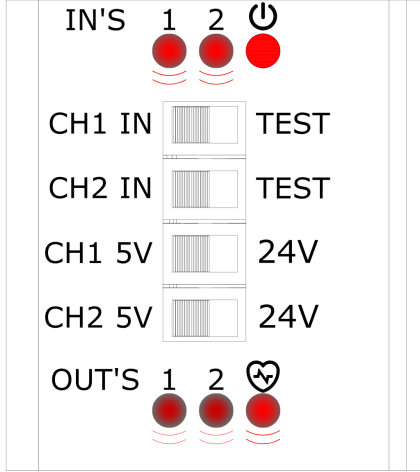
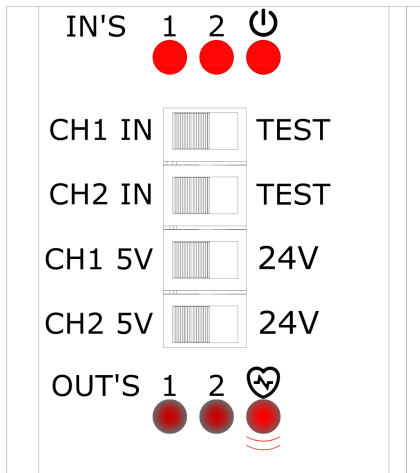
Switch Settings	Description
640/1000Hz	Select the switch position based on the application and the range of frequencies supplied by the sensor.
NORM/TEST	NORM: Set in this position for normal operation.
	TEST: Set when checking for proper hardware operation. When TEST mode is selected the respective OUTPUT LED will light if the hardware is working correctly (The INPUT LED will not light unless an actual input is being received). At the teach pendant check the variables shown in the software section to see if the controller software is correctly receiving the test input. The total pulse count should be increasing. If the "Threshold" system variable is more than 17 then the frequency reading will be zero. The frequency reading is typically 19 Hertz ± 2 .
Note If the switch is left in the TEST mode, the HSI will not provide readings of the frequency input.	

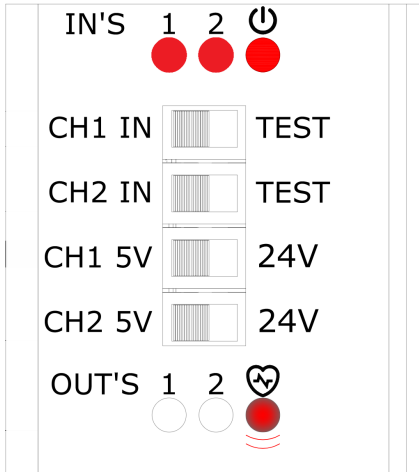
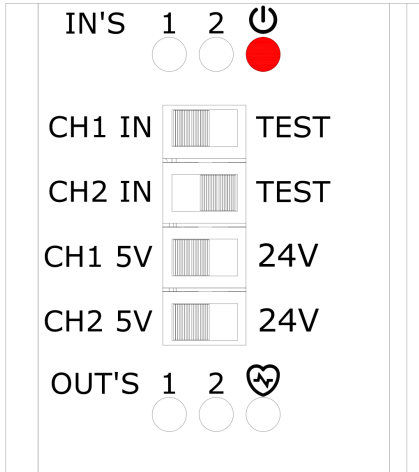
5.3 LED DESCRIPTIONS AND TROUBLESHOOTING INFORMATION

Table 5.3 (a) details the HSI Module LEDs and provides further troubleshooting information.

Table 5.3 (a) HSI Module LEDs - EE-8836-703-000

Operating State	LED STATUS	LED DESCRIPTION	OPERATING FAULT/CONDITION
Normal Operation Mode: IN Input X: No Input Signal	 Figure 5.3 (a) Normal Operation - No Input	Power LED	ON: 24VDC Power supplied to unit. OFF: 24VDC power is insufficient or missing: 1. Check Fuse 1 for continuity. 2. Verify cabling. 3. Replace the CPU cable. 4. Replace the HSI module.
		Heartbeat LED	Normal operation is ~24 Hz. Approximately 50% brightness. Units test clock functions properly.

Operating State	LED STATUS	LED DESCRIPTION	OPERATING FAULT/CONDITION
Normal Operation Mode: TEST Input X: Not applicable	 Figure 5.3 (b) Test Mode	OUT'S LED	Normal operation is ~24 Hz. Approximately 10% brightness. Unit's repeater circuit functions properly. Verify system variable changes with respect to the input signal. See Table 3.1. Replace HSI module if not flashing.
Normal Operation Mode: IN Input X: Frequency < 120Hz	 Figure 5.3 (c) Normal Operation - Slow Frequency	IN'S LED OUT'S LED	The IN LEDs will flash at the respective input frequency. Verify input signal to module. Approximately 10% to 20% brightness. Same Freq as IN LEDs. Device is operating properly. Verify system variable changes with respect to the input signal. See Table 3.1.
Normal Operation Mode: IN Input X: Frequency > 120Hz	 Figure 5.3 (d) Normal Operation - Fast Frequency	IN'S LED OUT'S LED	IN LEDS will illuminate at a fraction of the total LED brightness dependent on current of the input signal and duty cycle. OUT LEDS will illuminate at a fraction of the total LED brightness dependent on the input frequency. Verify system variable changes with respect to the input signal. See Table 3.1.

Operating State	LED STATUS	LED DESCRIPTION	OPERATING FAULT/CONDITION
Fault Condition Mode: IN Input X: Frequency > 1/TL Hz or Held High	 Figure 5.3 (e) Fault Condition - 1 Chanel 1 & 2 faulted shown.	IN'S LED IN LEDS will illuminate at a fraction of the total LED brightness dependent on current of the input signal and duty cycle.	OUT'S LED OUT'S LEDS "OFF" or not pulsing If frequency is greater then 1/TL the monostable will be held low. 1. Verify a pulse is being generated from the input sensor/device. 2. Ensure both input frequencies are less than 880 Hz. 3. Use the TEST mode to verify HSI module operation. 4. Replace the HSI module. 5. Reduce the input frequency if possible or replace the input sensor/device.
		Heartbeat LED Normal operation is ~24 Hz. Approximately 50% brightness. Unit tests clock functions properly. Replace HSI module if not flashing. Note The TEST function will only work if the heartbeat LED is pulsing.	
Fault Condition Mode: TEST Input X: Not applicable	 Figure 5.3 (f) Fault Condition - 2 Heartbeat shown faulted.		

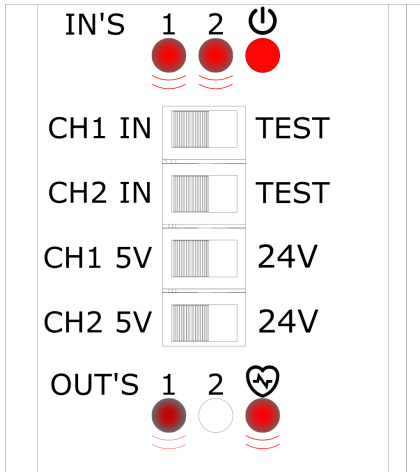
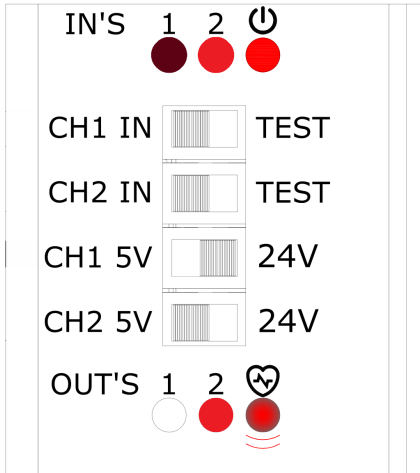
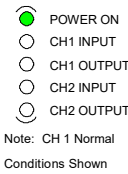
Operating State	LED STATUS	LED DESCRIPTION	OPERATING FAULT/CONDITION
Fault Condition Mode: IN Input X: Frequency < 120Hz	 Figure 5.3 (g) Fault Condition - 3 Chanel 2 fault shown.	IN'S LED	The IN LEDs will flash at the respective input frequency. Verify input signal to module.
		OUT'S LED	Approximately 10-20% brightness. Same Freq as IN's. Verify system variable changes with respect to the input signal. See Table 3.1 . 1. If the IN'S LED is pulsing or dim, the respective output should be pulsing. 2. Use the TEST mode to verify HSI module operation. 3. Replace the HSI module.
Fault Condition Mode: IN Input X: Frequency < 1/TL	 Figure 5.3 (h) Fault Condition - 4 Chanel 1 fault shown.	IN'S LED	CH 1 - 24V setting with 5V input. LED will be dimly lit. CH 2 - 5V setting with 5V input. LED will be normal brightness WARNING : USING 5V SETTING WITH 24 INPUT COULD DAMAGE THE MODULE.
		OUT'S LED	CH 2 is operating correctly. CH 1 is faulted. If CH 1 IN LED pulse is dim, the respective OUT LED will NOT pulse. 1. Current is too low to trigger the optoisolator. 2. Use TEST mode to verify HSI module operation. 3. Verify correct input specifications listed in Table 2.2. 4. Replace the HSI module.

Table 5.3 (b) HSI Module LEDs - EE-1063-300, EE-1063-400 (Obsolete)

LED Label	CH1/CH2 Mode Switches	LED Status	LED Description
POWER ON	TEST or NORM	ON	Normal Condition: The interface module is supplied with 24 VDC power.
		OFF	Fault Condition: The interface module is not supplied with 24 VDC power. The cable to JRM32 (R-J3) or JRL5 (R-J3iB, R-J3iC, R-30iA, R-30iB, and R-30iB Mate), on the CPU board, is not connected or the HSI Module has failed.

LED Label	CH1/CH2 Mode Switches	LED Status	LED Description	
CH1/CH2 INPUT	TEST or NORM	ON	Normal Condition: The actual Input signal is on but not pulsing or the input is pulsing at a high frequency.	
		FLASHING	Normal Condition: The Input signal is pulsing slowly (1 to about 20 Hertz). CH 2 INPUT OFF	
		OFF	Normal Condition: No input is connected or no pulses are being received. Fault Condition: If it is known that pulses are occurring, check the wiring of the input or try another HSI Module.	
CH1/CH2 OUTPUT	TEST	ON	Note: CH 1 Normal Conditions Shown	Normal Condition: The hardware is working properly.
CH1/CH2 OUTPUT	TEST	OFF		Fault Condition: If the POWER ON LED is illuminated then the module has failed. If this channel is being used, replace the module.
CH1/CH2 OUTPUT	NORM	ON	Note: CH 1 Normal Conditions Shown	Normal Condition: If the corresponding CH1 or CH2 Input LED is on or flashing then pulses are being sent at a high frequency. Fault Condition: If the corresponding CH1 or CH2 Input LED is off and the CH1 or Ch2 Output LED is on then the module has failed. Replace the module.
CH1/CH2 OUTPUT	NORM	FLASHING	Note: CH 1 Normal Conditions Shown	Normal Condition: Pulses are being sent slowly (1 to 20 Hertz)

LED Label	CH1/CH2 Mode Switches	LED Status	LED Description	
CH1/CH2 OUTPUT	NORM	OFF	 POWER ON CH1 INPUT CH1 OUTPUT CH2 INPUT CH2 OUTPUT Note: CH 1 Normal Conditions Shown	Normal Condition: If the corresponding CH1 or CH2 Input LED is off then the Module is OK. Fault Condition: If the corresponding CH1 or CH2 Input LED is on or flashing and the CH1 or CH2 Output LED is off then the module has failed. If the corresponding CH1 or CH2 Input LED is on it is still probably working properly (the input is on but not pulsing). If the corresponding CH1 or CH2 Input LED is on AND it is known that there are pulses occurring then it is likely that the module has failed.

5.4 INTERNAL JUMPERS

This section provides information on the internal jumpers of the HSI Module EE-1063-300 and EE-1063-400 which have been obsolete. For replacement of this component, please use EE-8836-703-000 else remove the two jumpers internal to the HSI module (Jumper HDR) for EE-1063-300 and EE-1063-400. See [Figure A \(a\)](#) for more details.

6 TIMING DIAGRAMS

The new HSI module (EE-8836-703-000) uses the activation of an optoisolator to transmit a sinking input to a monostable circuit. After the monostable is triggered by a following edge or when the diode is activated, a LOW pulse is triggered for time TL. TL varies based on a RC constant. [Figure 6 \(b\)](#) shows an example TEST input compared to its respective output. If module is operating correctly the status LEDs should look like the following:

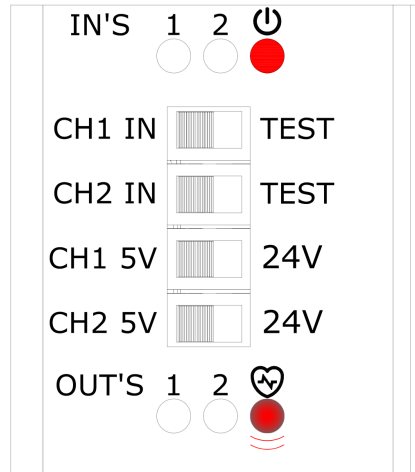


Figure 6 (a) Normal Operation - No Input

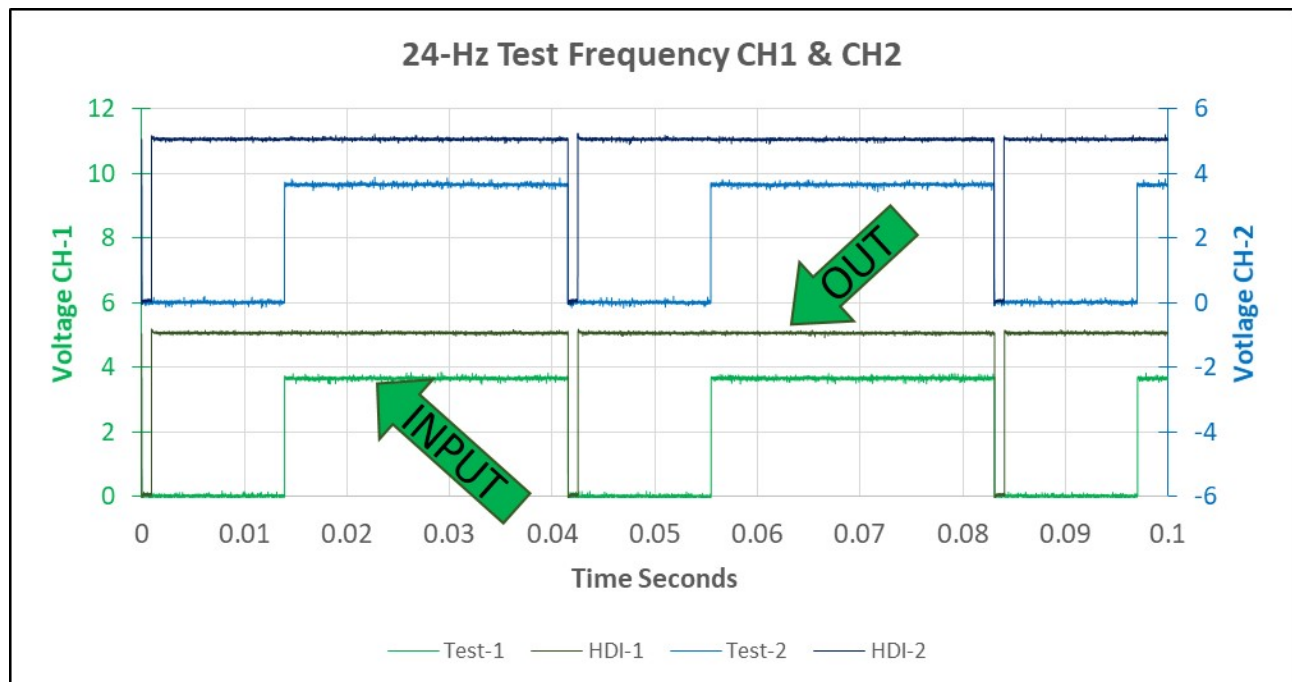


Figure 6 (b) 24Hz Test Frequency

Note

Please be aware there are two sources of delay.

1. Internal delay caused by the reaction of the monostable : [Figure 6 \(c\)](#).
2. Delay caused by the transition of the optoisolator: [Figure 6 \(g\)](#) or [Figure 6 \(f\)](#).

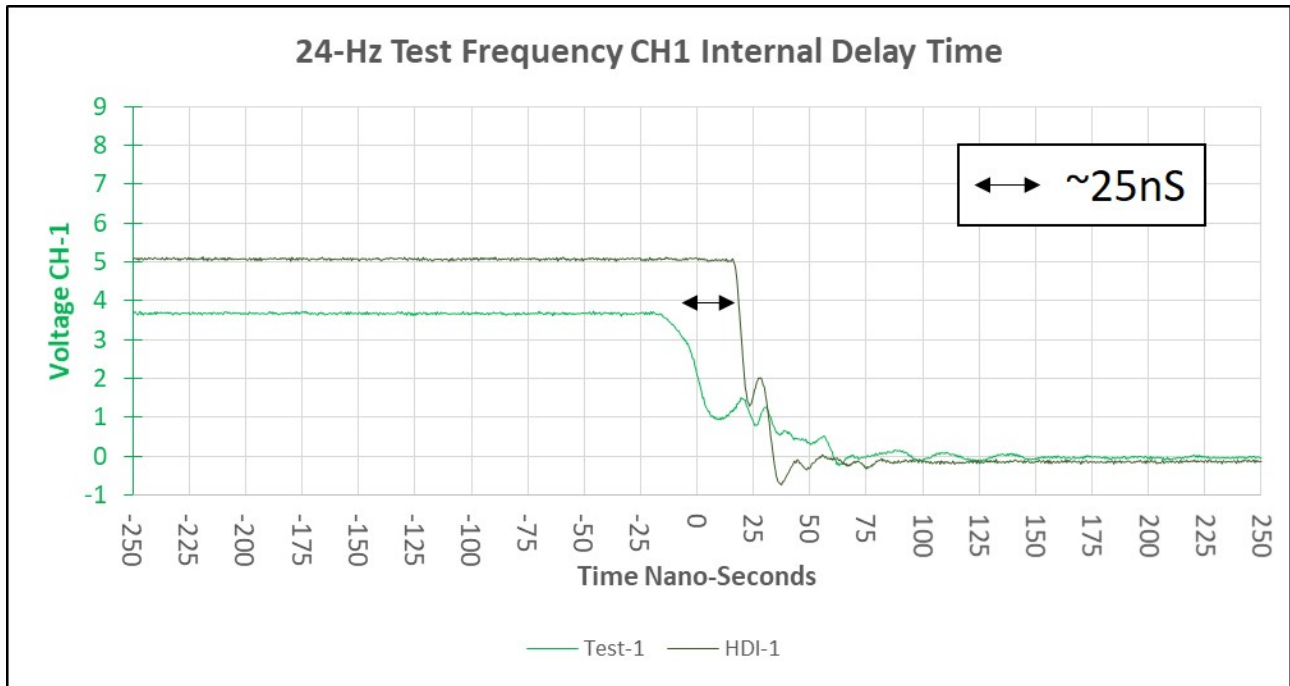


Figure 6 (c) 24Hz Internal Delay

The HSI modules do NOT share the same characteristics and could vary by 11%. Figure 6 (d) is an example of the pulse extension for the two channels. It is possible that even in the same model, the TL may be different. In this case, the maximum frequency of the module would be 1,030 Hz.

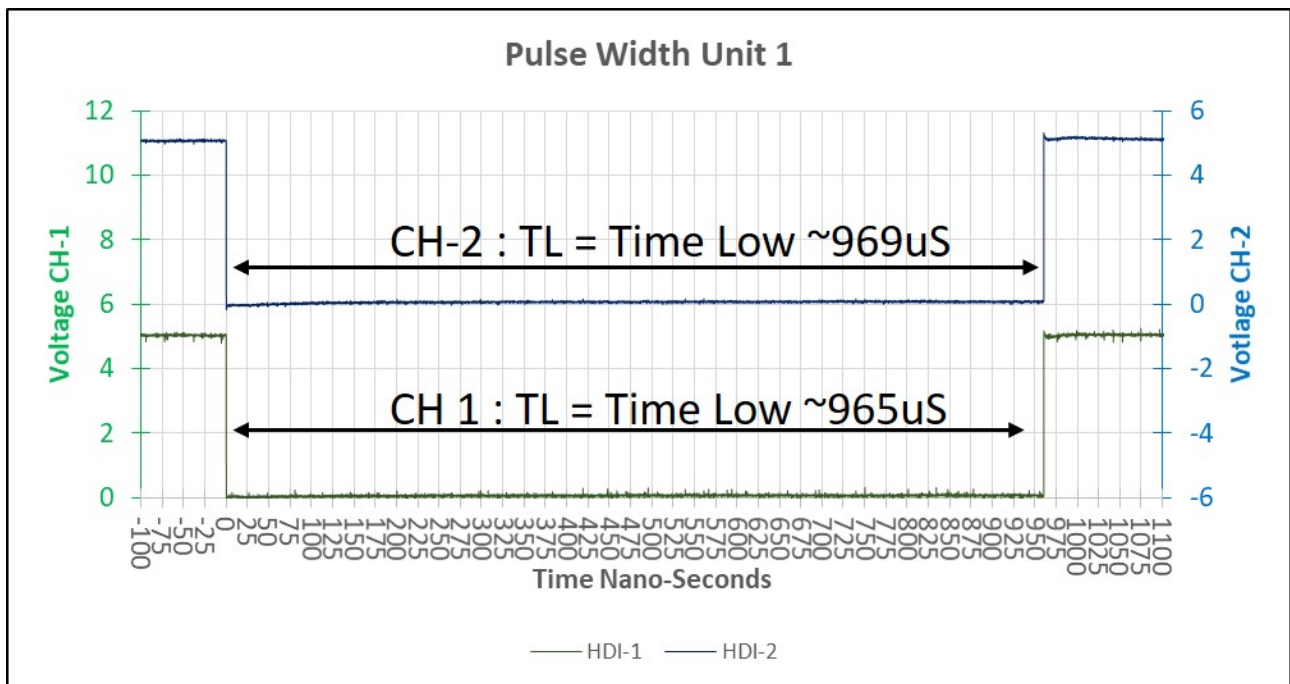


Figure 6 (d) Pulse Width Example

The Second function of the HSI module is to act as an interrupt. Figure 6 (e) is an example of a 150Hz input with a 55uS pulse or 0.8% duty cycle. The minimum pulse width is determined by the current flowing through the signal terminals. A larger current means a shorter pulse can be achieved.

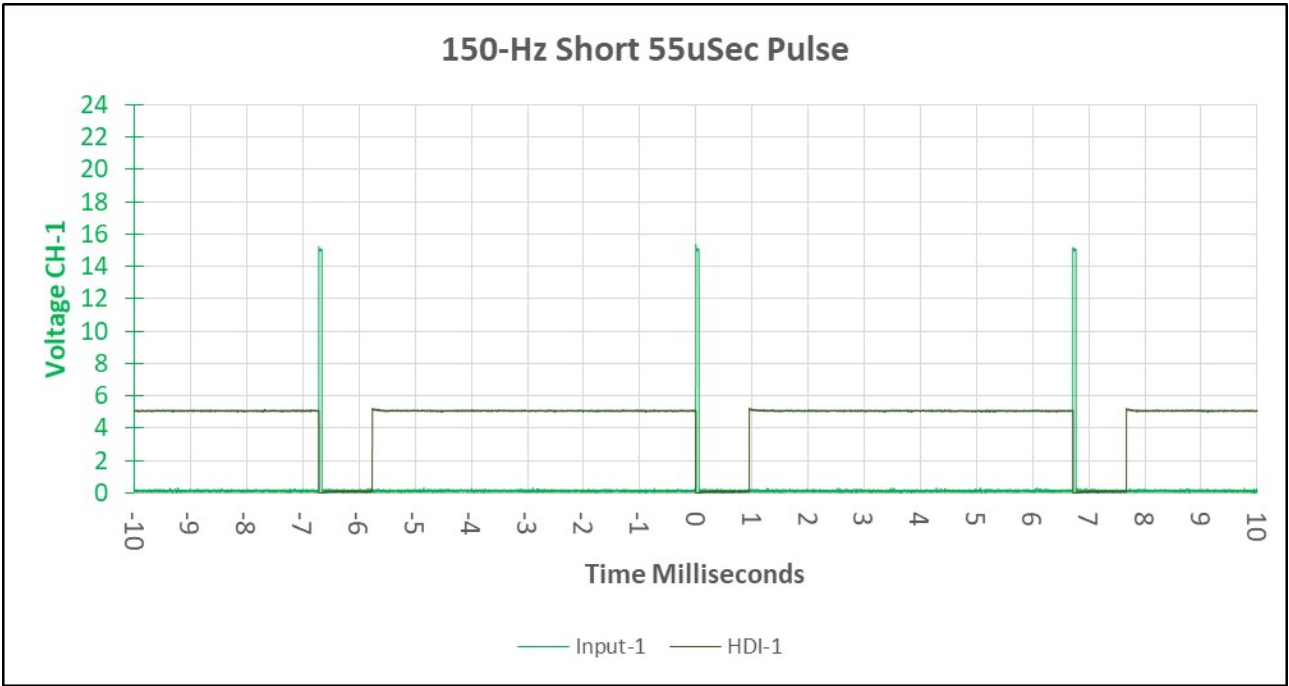


Figure 6 (e) Short Pulse Input

EE-8836-703-000 features a selection of a HIGH and LOW current setting. Ideally the input voltage is supplied from a 5VDC or 24VDC source. However, this does not restrict the input to 5V or 24V, rather it changes the response time as shown in [Figure 6 \(f\)](#) & [Figure 6 \(g\)](#). Under ideal conditions the module will react within 4uS, however line resistance, intrinsically safe barriers, or other components may delay the reaction time.

Note

Detailed data was collected from a sample module - device operation and characteristics may vary.

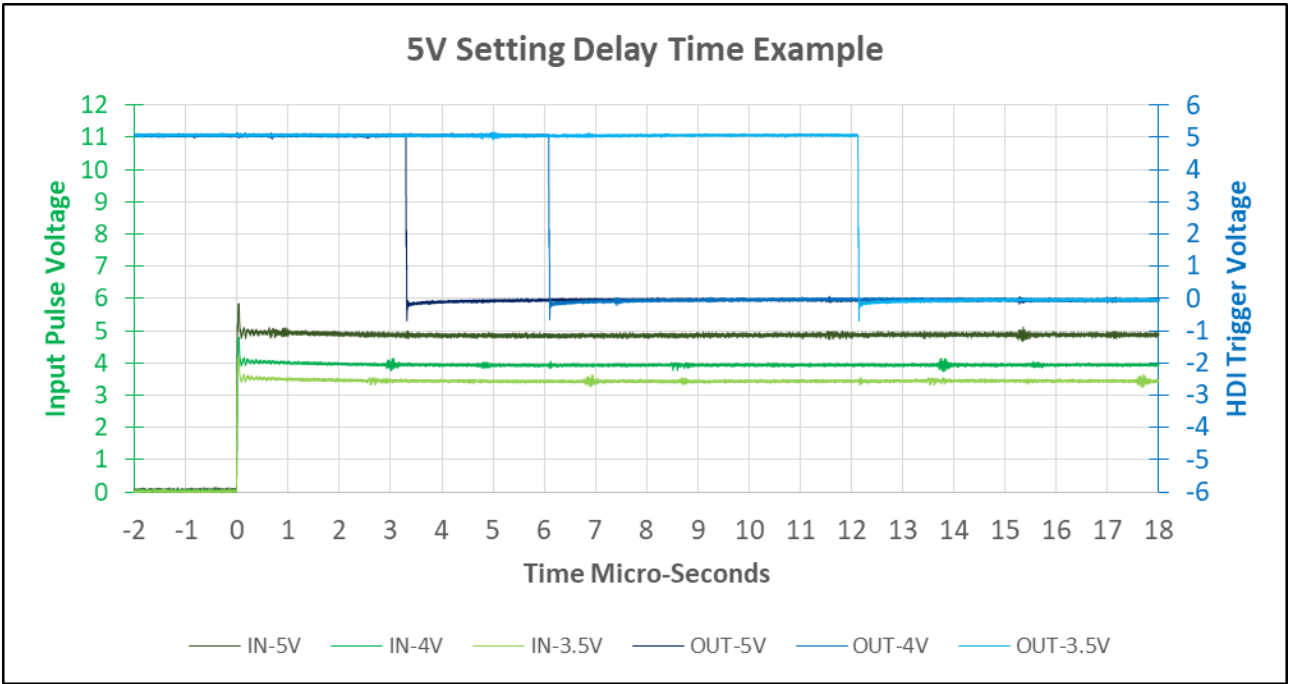


Figure 6 (f) 5V Setting Delay Time

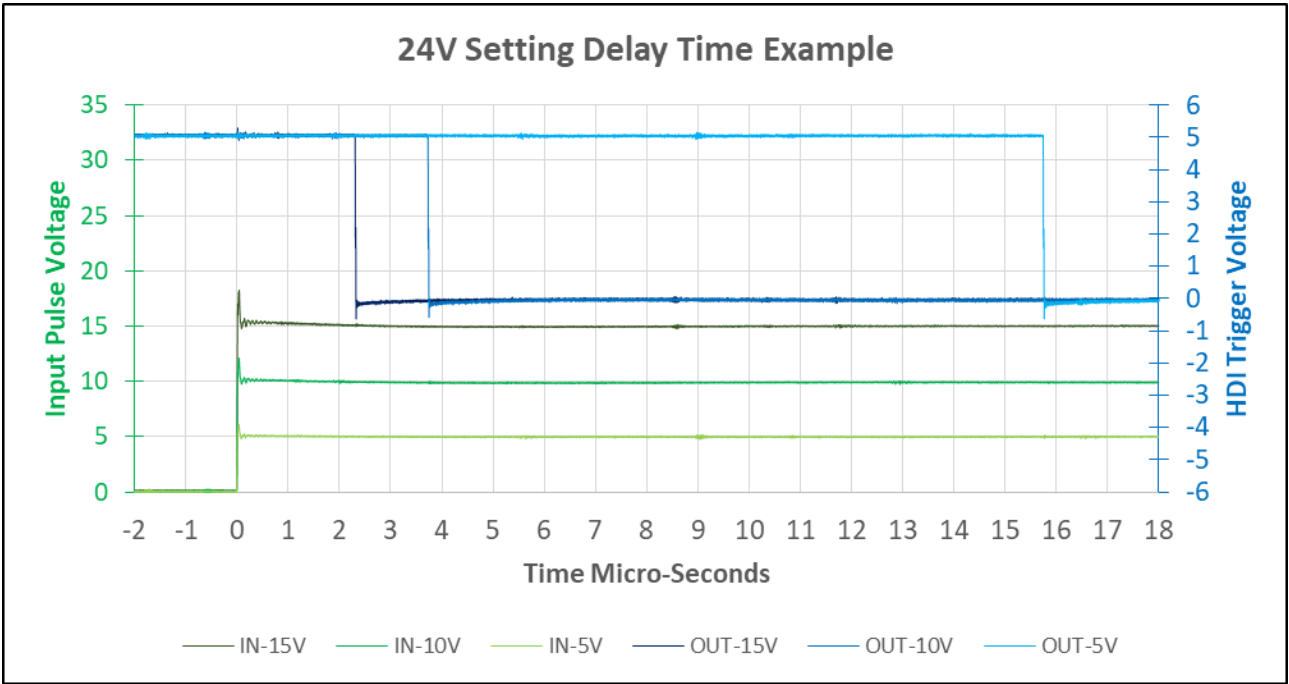
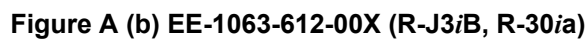


Figure 6 (g) 24V Setting Delay Time





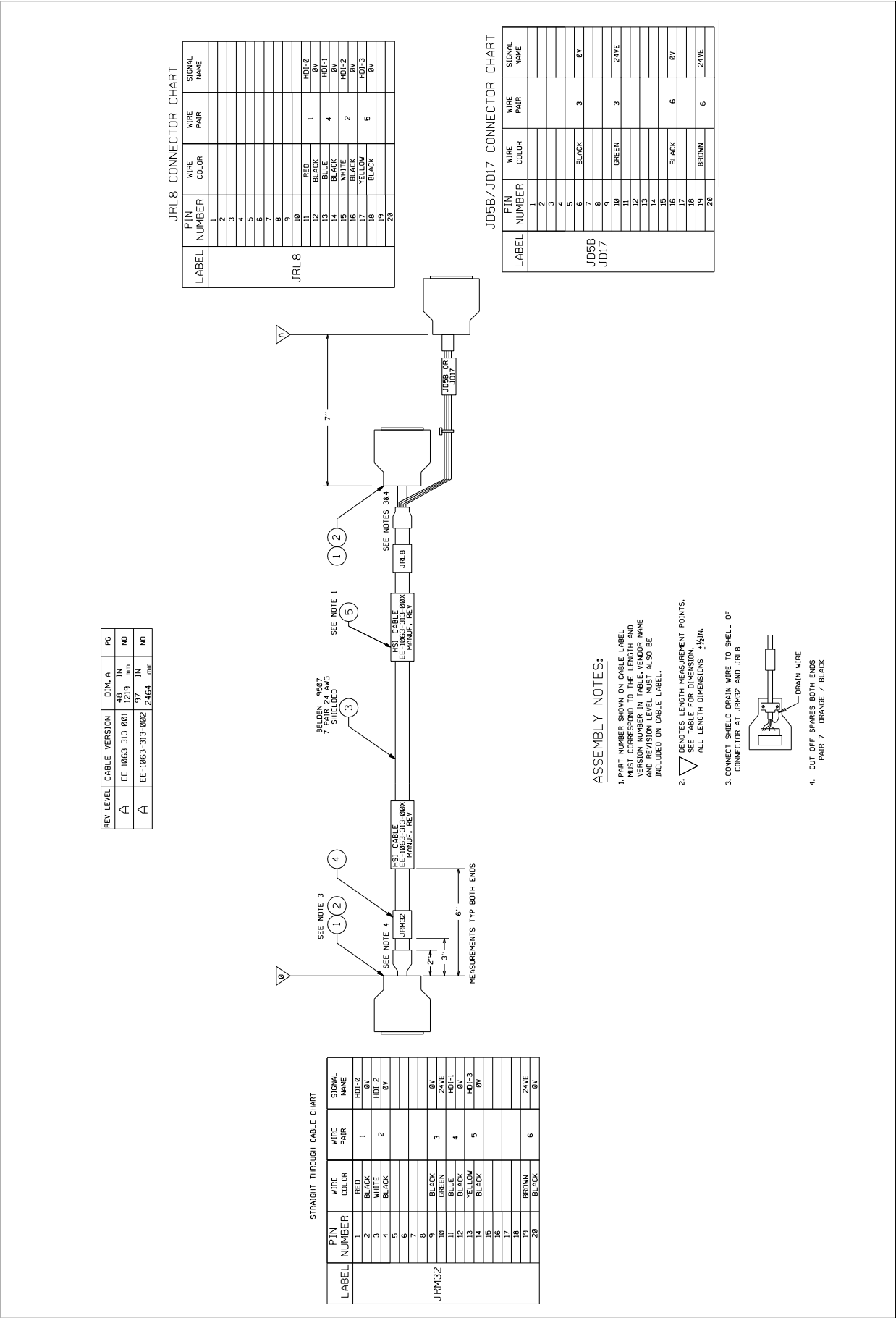


Figure A (c) EE-1063-313-001 (R30iB & R30iB PLUS)

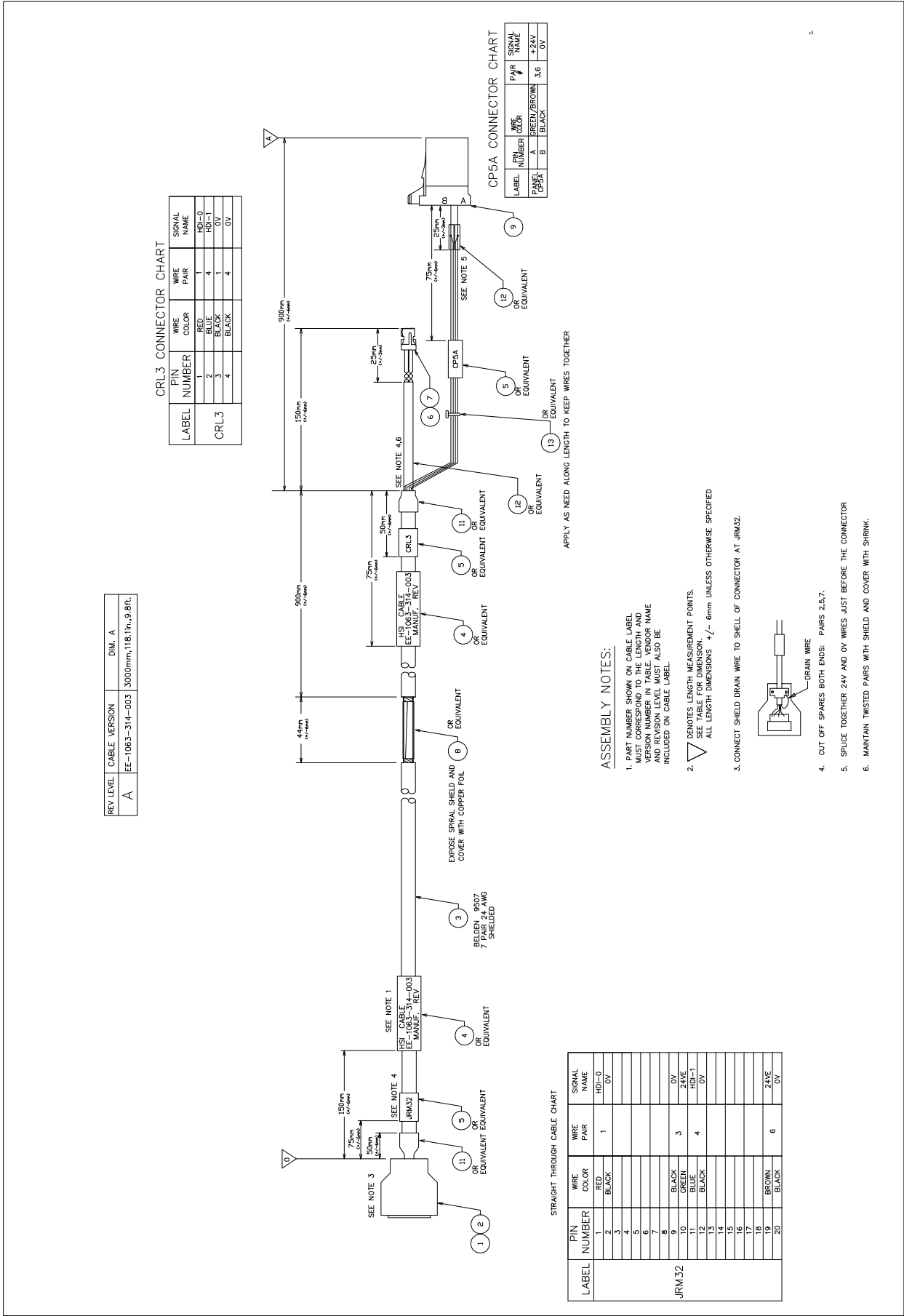


Figure A (d) EE-1063-314-001 (R30iB MATE & R30iB MATE PLUS)

REVISION RECORD

Edition	Date	Contents
REV F	Nov. 2020	Added new HSI Module EE-8836-703-000

